MEALWISE

A Smart Meal Expense & Eating-Habit Tracker

PRESENTED BY

Vimal K R

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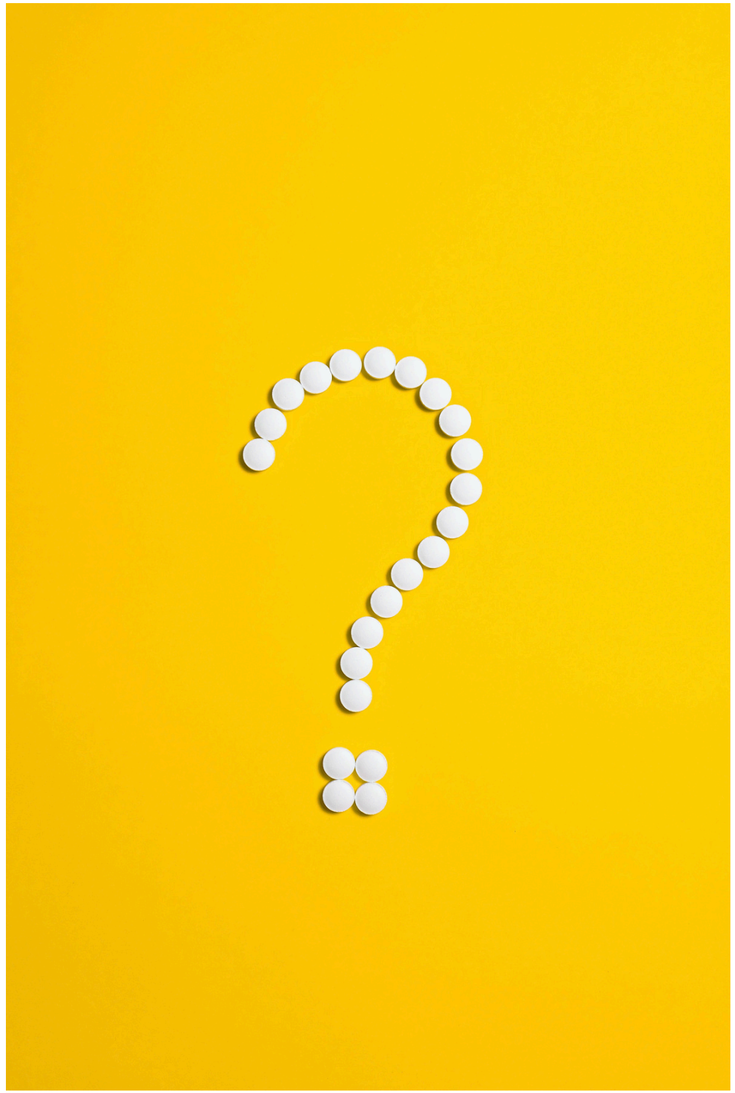
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Assistant Professor

INTRODUCTION

- Food spending is a daily, recurring expense, but it is rarely tracked with any real intent.
- Conventional budgeting apps record how much was spent, not what it was spent on.
- Diet and nutrition apps track health but leave cost out of the picture entirely.
- MealWise brings meal logging, budgeting, and eating-habit awareness together in one simple mobile app.



PROBLEM STATEMENT

To solve the difficulty users face in tracking their food expenses and understanding their eating habits, there is a need for a simple way to monitor food spending and healthy or unhealthy food choices in one place.

LITERATURE REVIEW

Name of Paper	Year	Methodology	Result	Limitation
Development of an Application for Expense Accounting	2020	Designed a mobile app that scans a receipt via phone camera, recognizes item text, and stores structured spending data.	Successfully implemented text recognition and expense-statistics storage for everyday purchases.	General-purpose expense tool; does not distinguish food quality or provide dietary insight.
Improving Heart Disease Risk Through Quality-Focused Diet Logging	2020	Built an app around the Healthy Heart Score; users log simplified food categories instead of full nutrition data. Tested with 38 participants over 5 weeks.	Simplified category-based logging significantly improved diet-quality scores and lowered predicted heart-disease risk.	Focused only on health scoring; the app has no budget or expense-tracking component.
Expense Tracking with Tesseract OCR v5: A Mobile Application Development	2023	Built a mobile app using Tesseract OCR v5 (LSTM) to extract expense line items from scanned grocery/market receipts; evaluated on 20 real receipts.	Achieved low Character/Word Error Rates for item extraction, enabling largely automated expense logging.	OCR accuracy depends on receipt image quality; captures amounts only, with no link to nutrition or health.



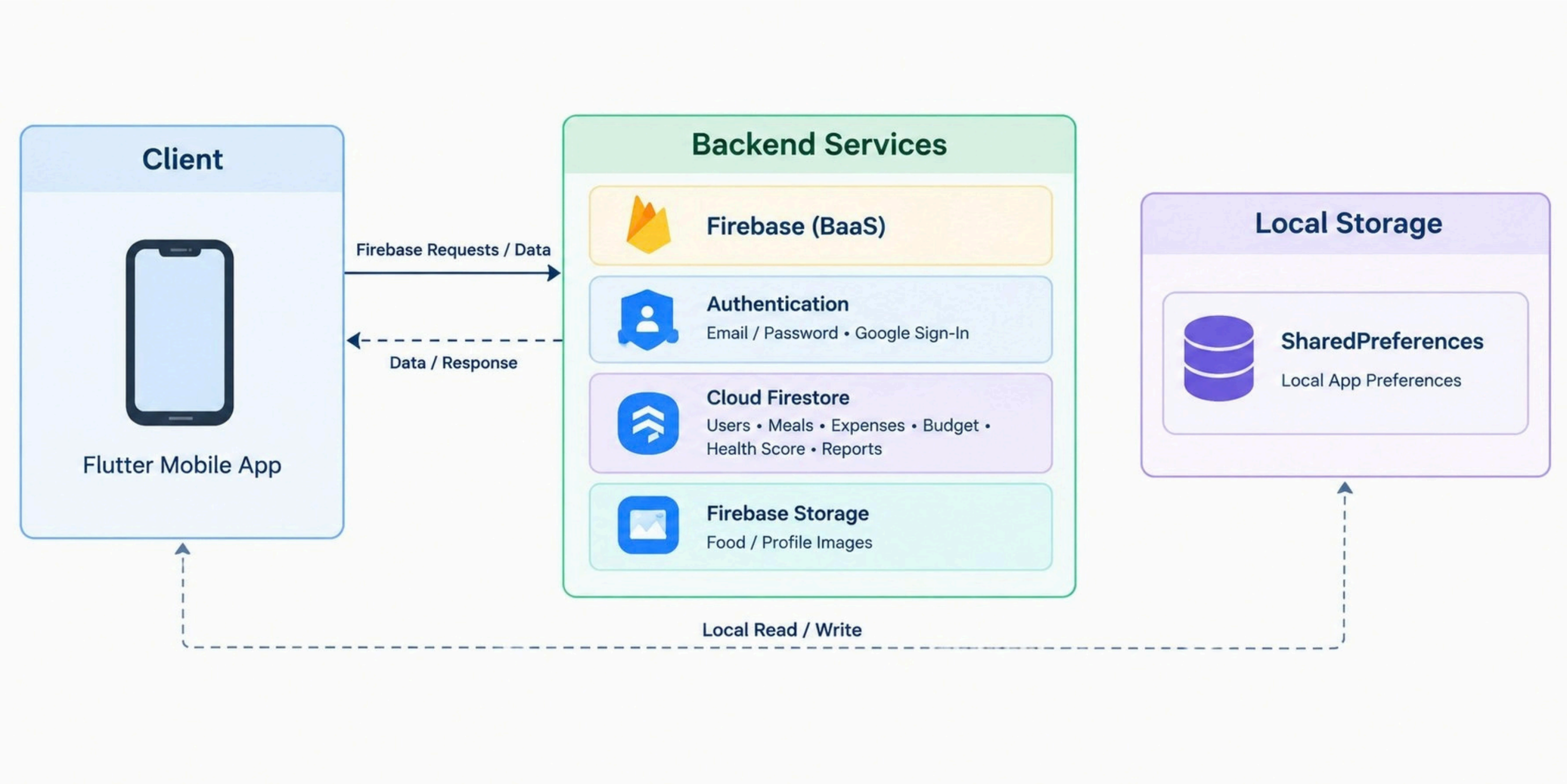
GAP IDENTIFIED

- Expense tracking apps record spend, but ignore what food was actually bought.
- Diet quality apps score eating habits, but leave cost out entirely.
- Detailed nutrition logging (calories, macros) is time consuming and hard to sustain daily.
- There is a need for a lightweight app that logs cost and food category together in a single step.

PROPOSED SOLUTION

- Log each meal with its cost and a simple food category in a single entry.
- Track a user-defined monthly food budget against actual spending in real time.
- Compute a Monthly Health Score from the health value of the categories logged.
- Provide a Calendar view and a Monthly Report to review spending and habits over time.
- Sync data securely to the cloud via Firebase so records persist across sessions and devices.

SYSTEM ARCHITECTURE



ARCHITECTURE OVERVIEW

- MealWise follows a client backend architecture.
- The Flutter mobile app acts as the client and handles the user interface.
- All core data operations login, saving meals, budgets, and reports are managed through Firebase, which acts as the backend service.
- Small on device settings are stored separately using local storage, so they load instantly without needing the internet.

ARCHITECTURE COMPONENTS

1. Client Layer

- Provides the user interface Home, Log Meal, Calendar, and Report screens.
- Sends requests to Firebase to save or fetch data.
- Displays the data and responses received back from Firebase.

2. Backend Layer

- Authentication: manages user login via Email/Password and Google Sign-In.
- Cloud Firestore: stores users, meals, expenses, budget, health score, and reports.
- Firebase Storage: stores food and profile images.

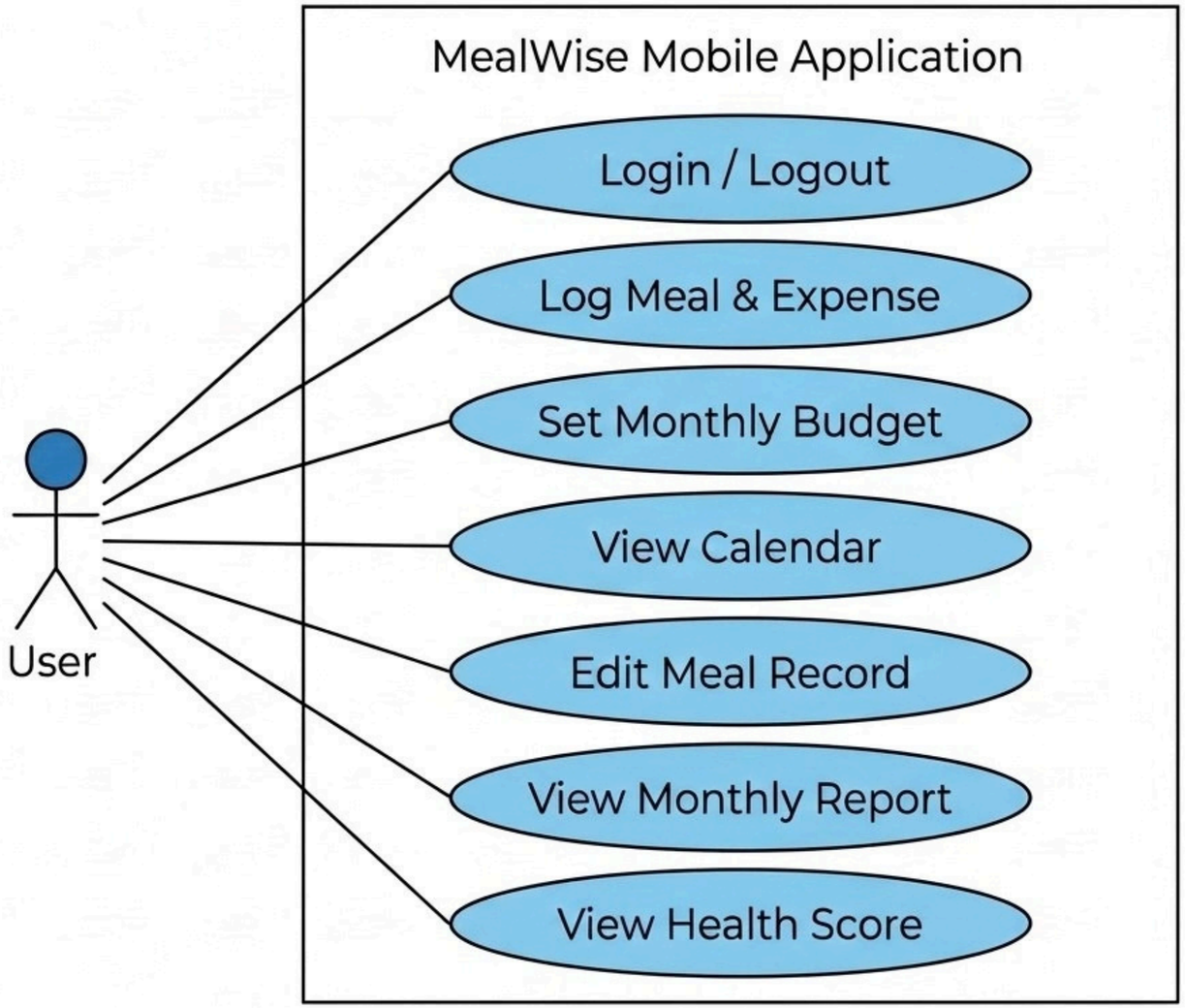
3. Local Storage Layer

- SharedPreferences stores local app settings and preferences (e.g. theme, onboarding status).
- Works directly on the device, independent of internet connectivity.

MODULE DESCRIPTION

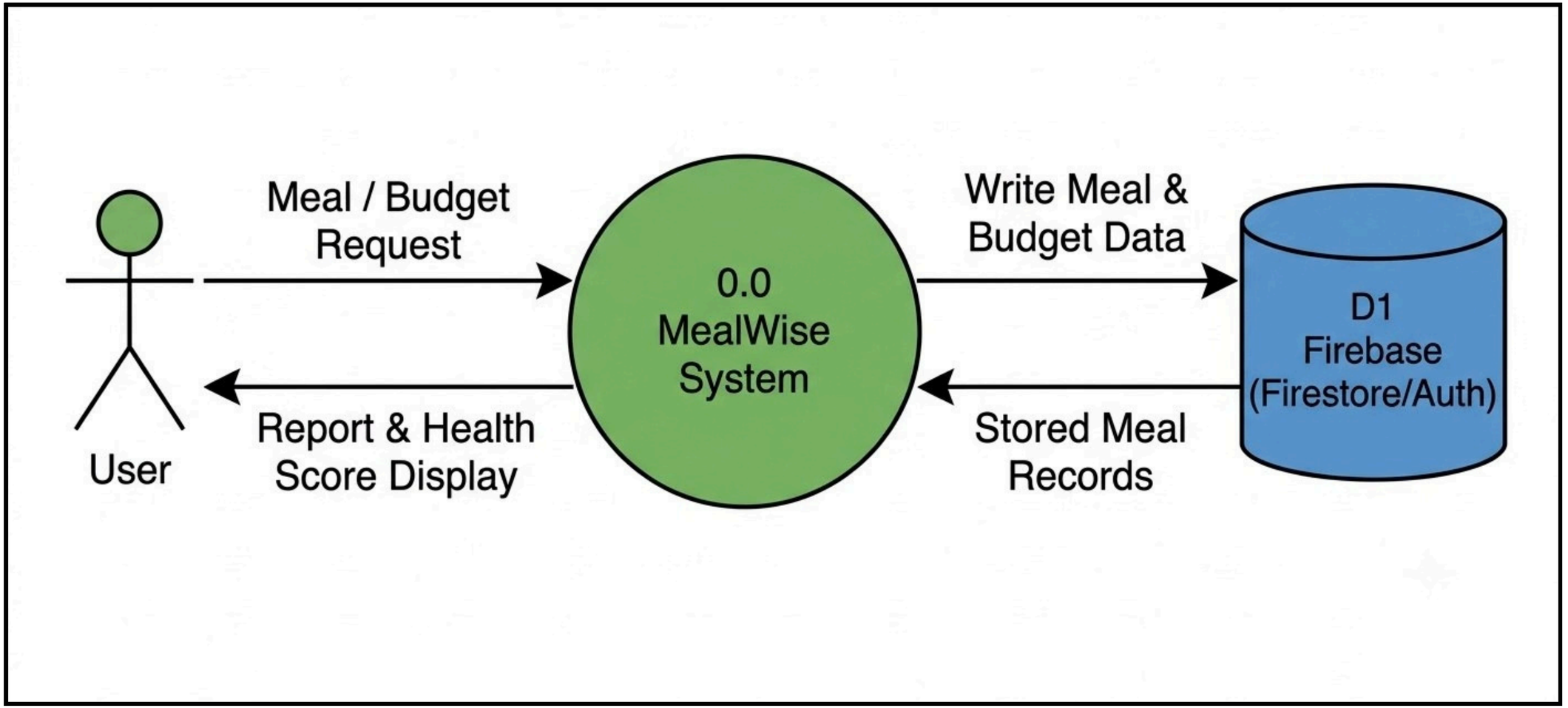
- **Frontend UI**: Cross-platform Flutter interface for meal logging, budget viewing, and reports.
- **Authentication Module**: Manages sign-up, login, and Google sign-in through Firebase Auth.
- **Meal Logging Module**: Captures meal type, name, cost, location, time, and notes.
- **Database Module**: Stores and streams meal records in real time via Cloud Firestore.
- **Local Storage Module**: Caches budget, name, and settings using SharedPreferences.
- **Reporting Module**: Aggregates meals into calendar views and monthly spending reports.
- **Health Score Module**: Converts logged food categories into a Monthly Health Score.

USE CASE DIAGRAM

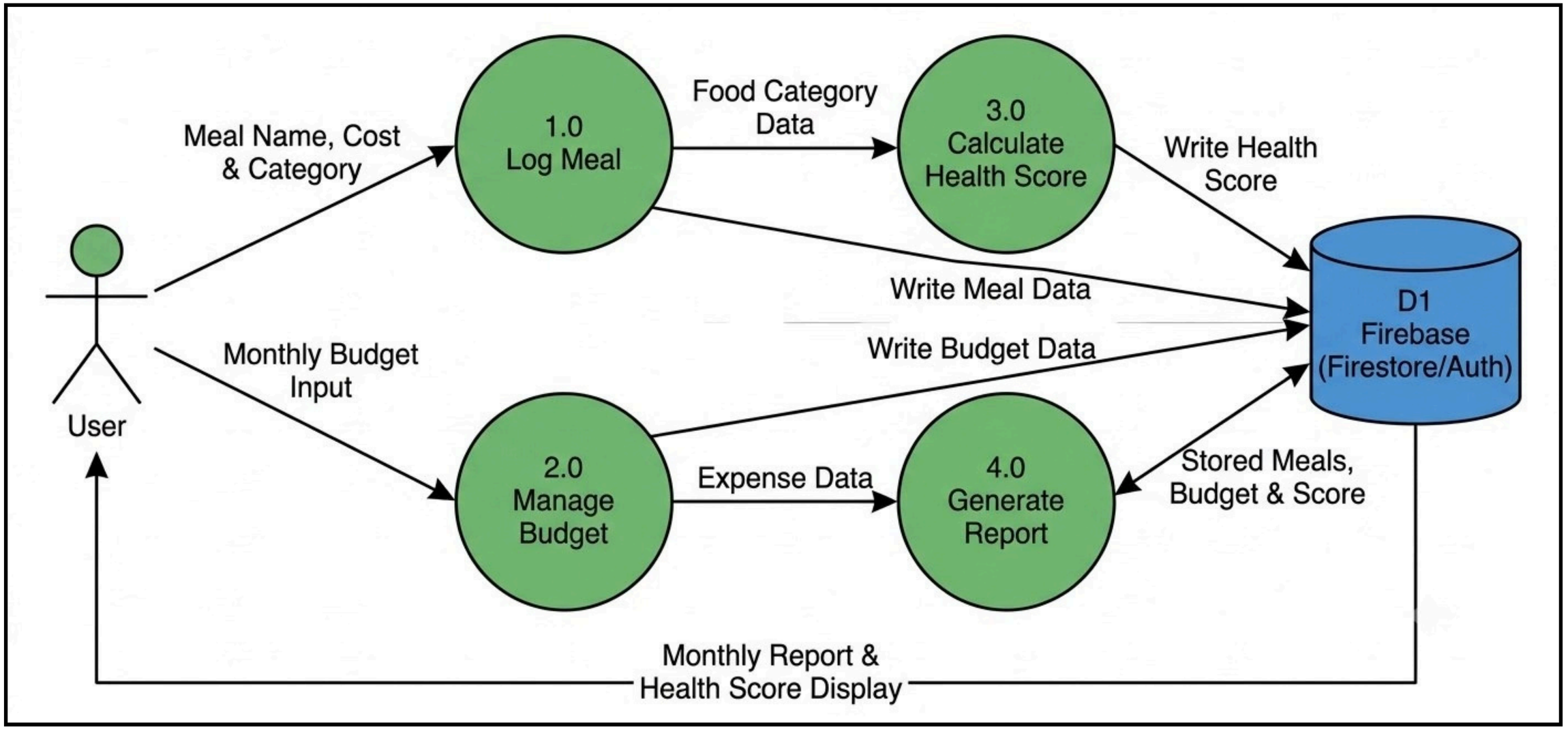


DATA FLOW DIAGRAM

LEVEL O


Meal / Budget
Request
Write Meal &
Budget Data
0.0
MealWise
System
D1
Firebase
(Firestore/Auth)
User
Report & Health
Score Display
Stored Meal
Records

LEVEL 1


Meal Name, Cost & Category
1.0 Log Meal
Food Category Data
3.0 Calculate Health Score
Write Health Score
Write Meal Data
D1 Firebase (Firestore/Auth)
User
Monthly Budget Input
Write Budget Data
2.0 Manage Budget
Expense Data
4.0 Generate Report
Stored Meals, Budget & Score
Monthly Report & Health Score Display

DATABASE REQUIREMENTS

- Database Type: Firebase Cloud Firestore a document-based NoSQL database.
- Stores each user's meal records under users/{uid}/meals for clean data isolation.
- Access is controlled through Firebase Security Rules scoped to the authenticated user.
- Supports real-time listeners so the UI updates instantly as data changes.

SOFTWARE REQUIREMENT SPECIFICATION

USER INTERFACE

- The app provides a clean, mobile-first interface built with Flutter.
- Styled using Google Fonts for consistent, readable typography.
- Budget and spending are shown visually on the Home screen with a progress indicator.
- Simple forms with validation guide users through logging a meal.

SOFTWARE REQUIREMENTS

- **Operating System**: Android / iOS / Windows / macOS / Linux (via Flutter)
- **Frontend Framework**: Flutter (Dart)
- **Backend & Database**: Firebase — Authentication, Cloud Firestore, Storage
- **Local Storage**: SharedPreferences
- **Fonts & Styling**: Google Fonts, Material theming
- **Package Manager**: pub (Dart / Flutter)
- **IDE**: Android Studio / Visual Studio Code

HARDWARE REQUIREMENTS

- A smartphone, tablet, or computer capable of running the Flutter app or a modern web browser.
- Active internet connection for account sign-in and real-time cloud sync with Firebase.
- Minimum 2 GB RAM recommended for smooth app performance on mobile devices.

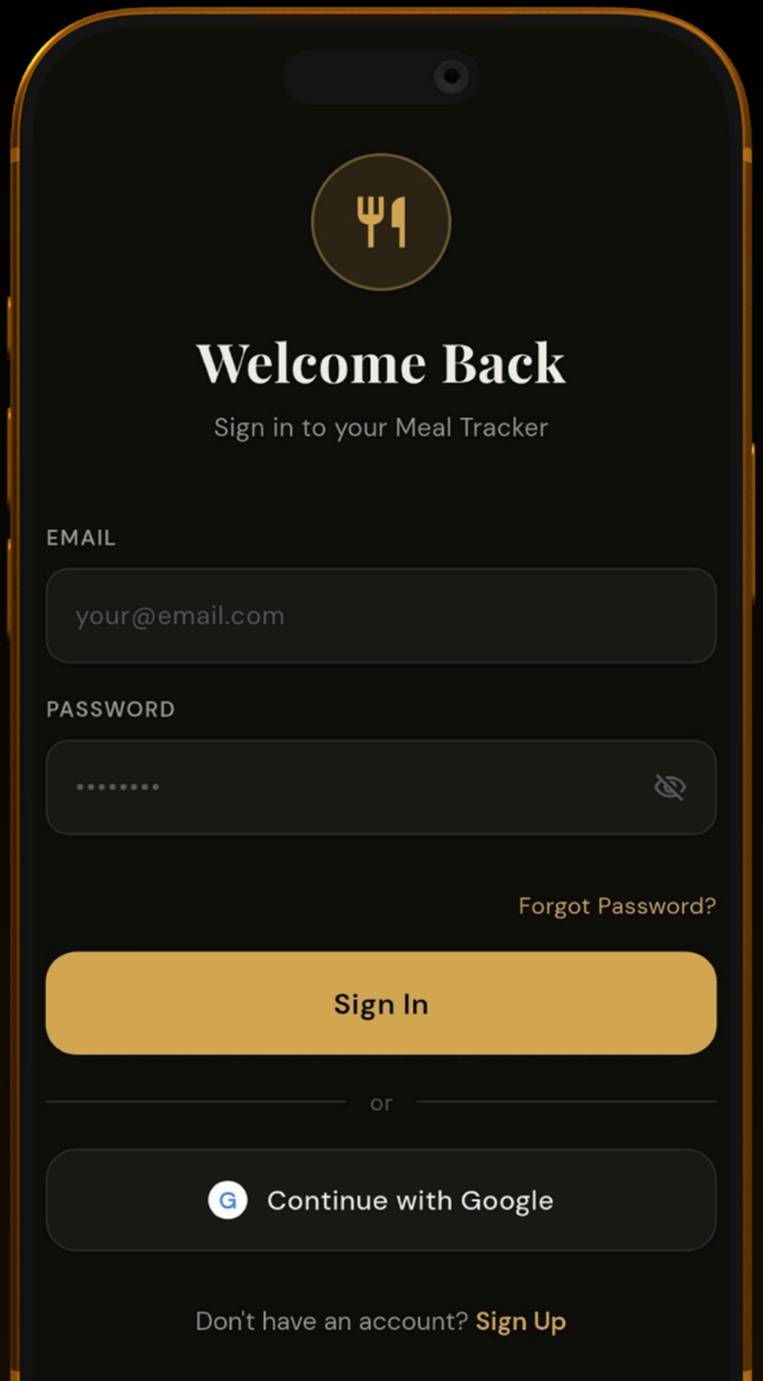
FUNCTIONAL REQUIREMENTS

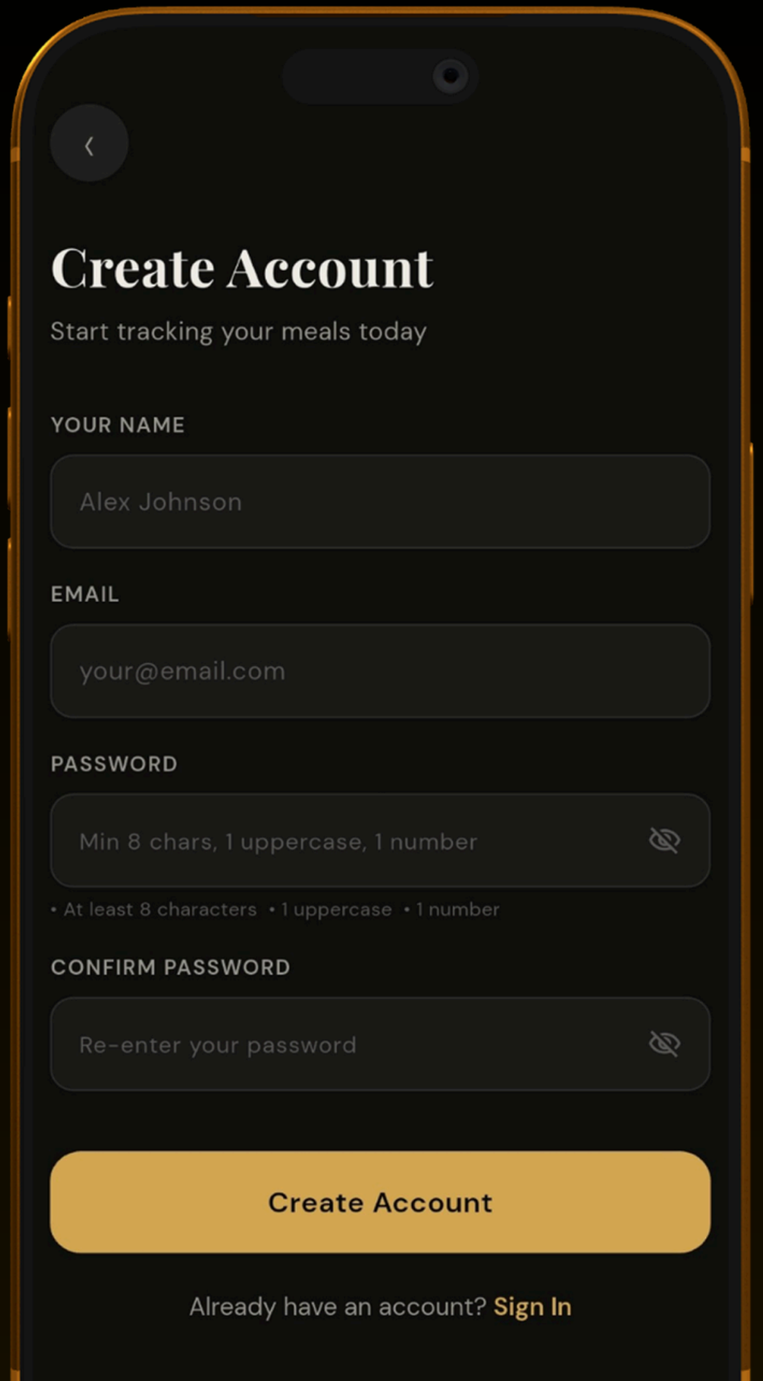
- **Login/Signup** : secure user authentication.
- **Log Meal** : enter meal name, cost, date, and category.
- **Set Budget** : define and track a monthly food budget.
- **Calendar View** : browse past meals by date.
- **Edit/Delete Meal** : update or remove a logged entry.
- **Monthly Report** : view spending and category breakdown.
- **Health Score** : auto-calculated from food categories logged.
- **Budget Alerts** : notify when nearing/exceeding budget.

NON-FUNCTIONAL REQUIREMENTS

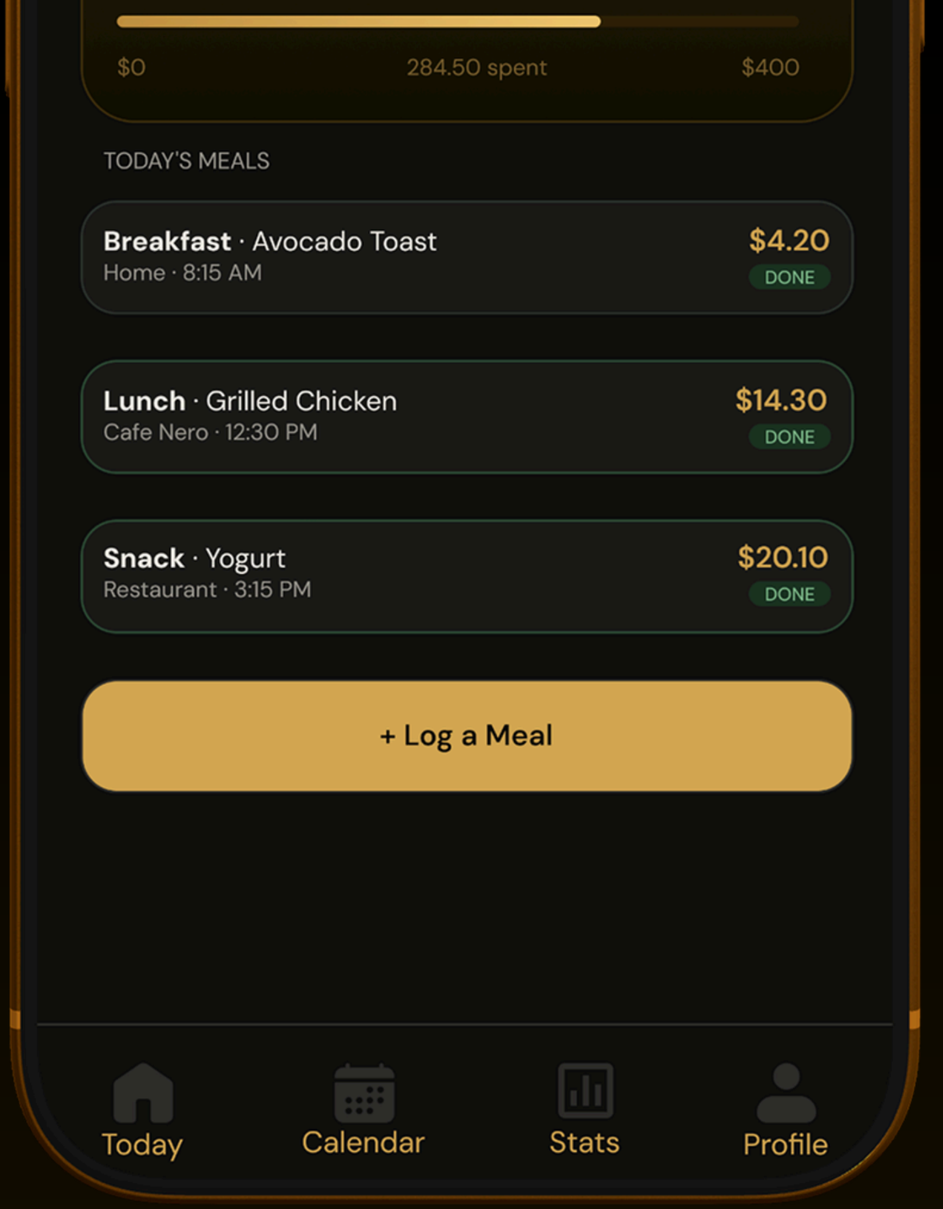
- **Performance**: Screens load and update quickly via Firestore's real-time listeners.
- **Security**: User data is isolated per account using Firebase Authentication and security rules.
- **Scalability**: Firestore's NoSQL structure supports growth in users and meal records.
- **Usability**: Simple, form-based meal logging requires minimal effort.
- **Reliability**: Cloud sync ensures data is not lost if the device is changed.
- **Responsiveness**: The UI adapts cleanly across phone, tablet, and desktop screens.

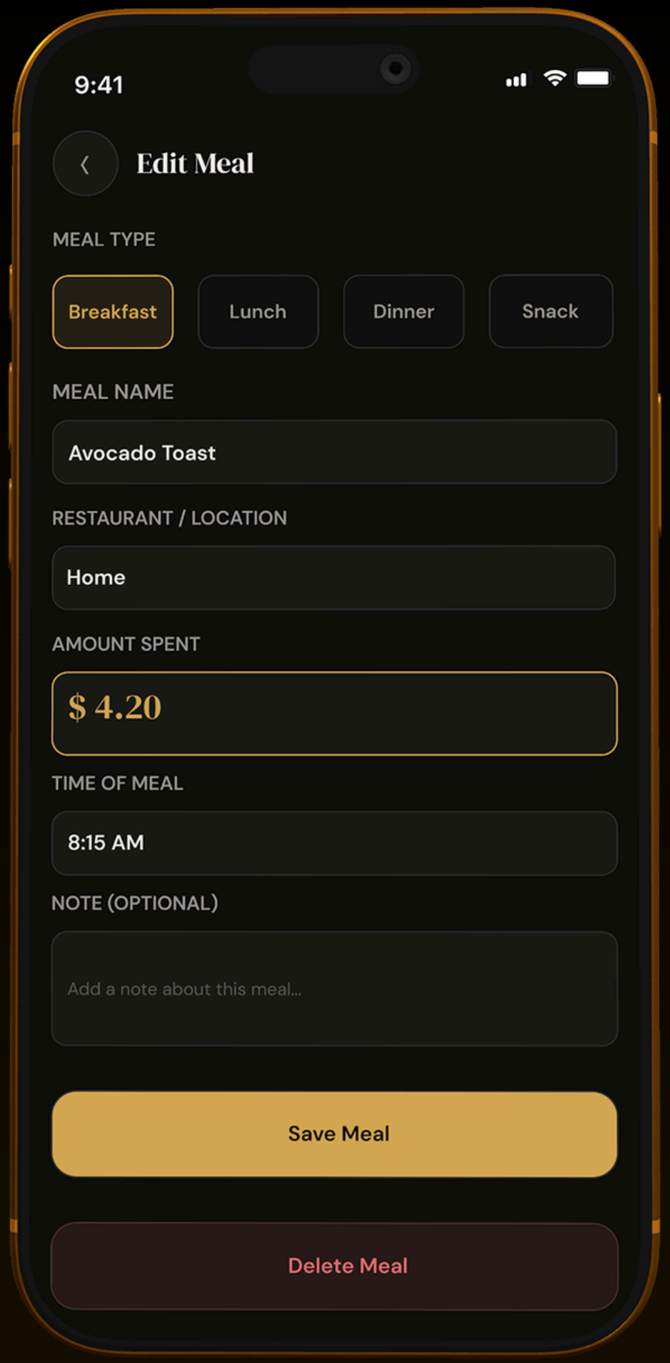
RESULTS

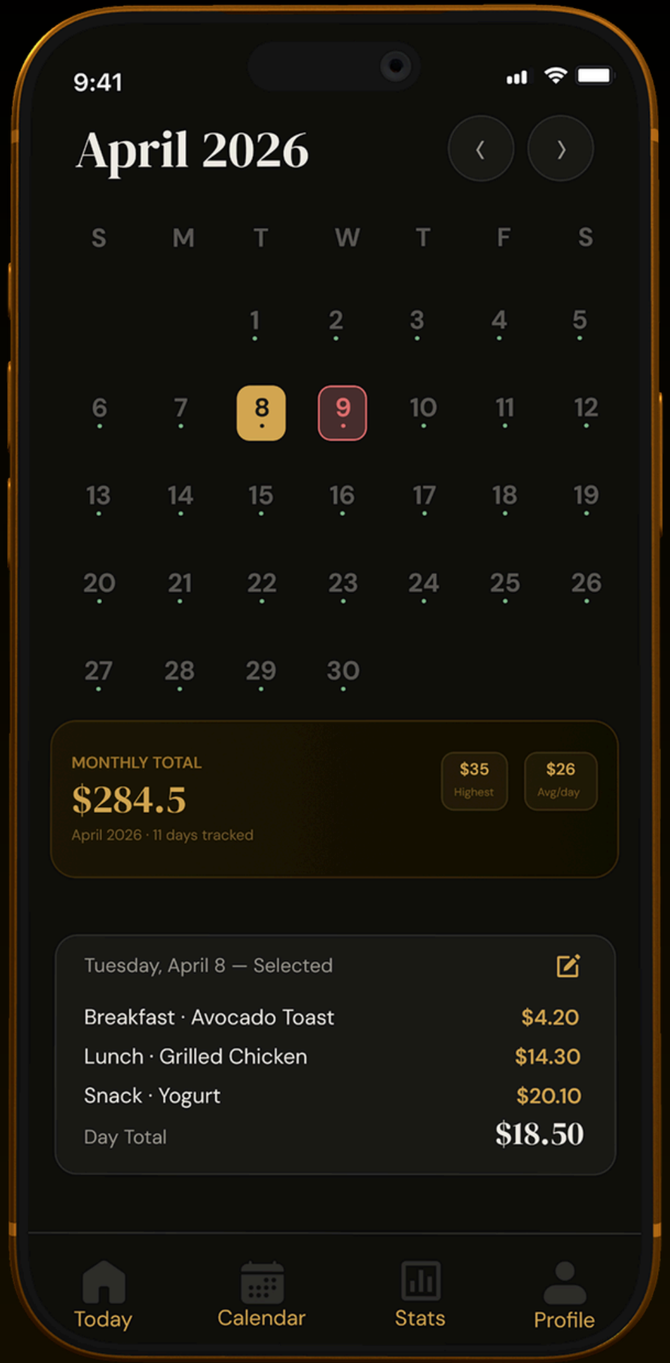

Welcome Back
Sign in to your Meal Tracker
EMAIL
your@email.com
PASSWORD
Forgot Password?
Sign In
or
Continue with Google
Don't have an account? Sign Up

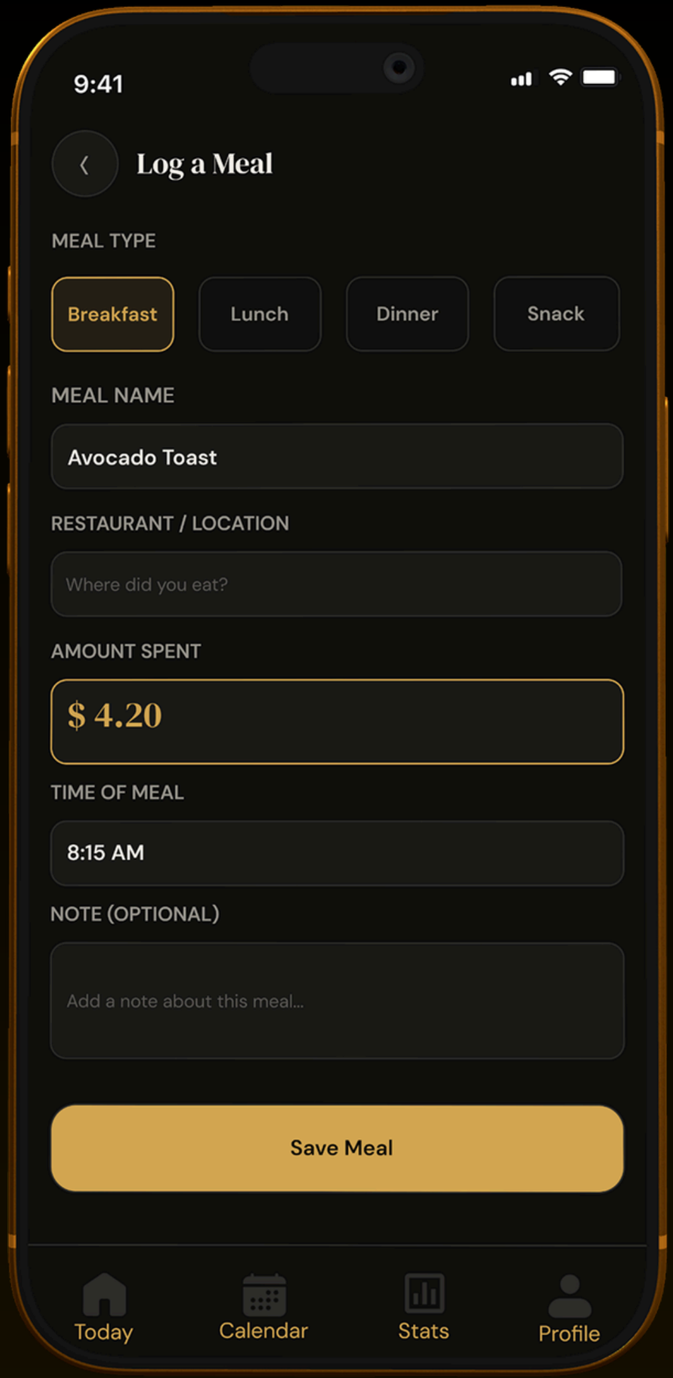

Create Account
Start tracking your meals today
YOUR NAME
Alex Johnson
EMAIL
your@email.com
PASSWORD
Min 8 chars, 1 uppercase, 1 number
• At least 8 characters • 1 uppercase • 1 number
CONFIRM PASSWORD
Re-enter your password
Create Account
Already have an account? Sign In


MealWise
Good morning
Alex Johnson
Tuesday, April 8 · 2026
TODAY'S SPEND
$38.6
April budget · $284 of $400 used
$0
284.50 spent
$400
TODAY'S MEALS
Breakfast · Avocado Toast
Home · 8:15 AM
$4.20
DONE
Lunch · Grilled Chicken
Cafe Nero · 12:30 PM
$14.30
DONE


$0
284.50 spent
$400
TODAY'S MEALS
Breakfast · Avocado Toast
Home · 8:15 AM
$4.20
DONE
Lunch · Grilled Chicken
Cafe Nero · 12:30 PM
$14.30
DONE
Snack · Yogurt
Restaurant · 3:15 PM
$20.10
DONE
+ Log a Meal
Today
Calendar
Stats
Profile


9:41
Edit Meal
MEAL TYPE
Breakfast
Lunch
Dinner
Snack
MEAL NAME
Avocado Toast
RESTAURANT / LOCATION
Home
AMOUNT SPENT
$ 4.20
TIME OF MEAL
8:15 AM
NOTE (OPTIONAL)
Add a note about this meal...
Save Meal
Delete Meal


9:41
April 2026
S M T W T F S
1 2 3 4 5
6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30
MONTHLY TOTAL
$284.5
April 2026 · 11 days tracked
$35
Highest
$26
Avg/day
Tuesday, April 8 — Selected
Breakfast · Avocado Toast $4.20
Lunch · Grilled Chicken $14.30
Snack · Yogurt $20.10
Day Total $18.50
Today
Calendar
Stats
Profile


9:41
Log a Meal
MEAL TYPE
Breakfast
Lunch
Dinner
Snack
MEAL NAME
Avocado Toast
RESTAURANT / LOCATION
Where did you eat?
AMOUNT SPENT
$ 4.20
TIME OF MEAL
8:15 AM
NOTE (OPTIONAL)
Add a note about this meal...
Save Meal
Today
Calendar
Stats
Profile

9:41

Monthly Report

April 2026 · 11 days tracked

TODAY'S SPEND

$284.50

$400.00 monthly budget

$0 — $115.50 remaining — $400

$35.80 Highest Day Apr 10	**$9.20** Lowest Day Apr 2
$25.86 Daily Average	**33** Meals Logged

SPEND BY MEAL TYPE

Breakfast	$80.20
Lunch	$128.10
Dinner	$59.80
Snacks	$16.40

Today · Calendar · Stats · Profile

9:41

Profile

Edit



Alex Johnson

alex@email.com

MONTHLY BUDGET

$400.00 Change

Resets on the 1st of each month

PREFERENCES

- **Meal Reminders** — Get reminded to log breakfast, lunch & dinner
- **Show history** — View your past meals and spending
- **Insights** — View Monthly Stats
- **Data & Reset** — Clear or reset your data anytime

Sign Out

Today · Calendar · Stats · Profile

GANTT CHART


No.
Task / Activity
Week 1
Week 2
Week 3
Week 4
Week 5
Week 6
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
Week 13
Week 14
1 Project planning & requirement analysis
2 Literature review & gap identification
3 Problem definition, objectives & scope
4 UI/UX design and screen planning
5 Flutter project setup & core UI development
6 Firebase Authentication
7 Cloud Firestore database integration
8 Meal logging module
9 Budget & spending tracking
10 Calendar/history and monthly reports
11 Integration, validation & bug fixing
12 Monthly Health Score implementation
13 Final testing, refinement & validation
14 Documentation, presentation & demo preparation
Completed
Planned / Pending
Milestone
Current Progress – Core Features Completed (Week 10)
Project Completion / Final Demo (Week 14)

CONCLUSION

- MealWise brings meal cost tracking and eating habit awareness into a single mobile app.
- Combines Flutter's cross platform UI with Firebase's real time cloud backend.
- Addresses the gap between expense trackers and diet quality apps identified in the literature.
- Lays the foundation for the planned Monthly Health Score to encourage healthier choices.

REFERENCES

- D. E. Yurochkin, A. A. Horoshiy, and S. A. Karpukhin, “Development of an application for expense accounting,” in Proc. 2021 IEEE Conf. Russian Young Researchers in Electrical and Electronic Engineering (ElConRus), St. Petersburg and Moscow, Russia, Jan. 2021, doi: 10.1109/ElConRus51938.2021.9396201.

- B. C. Kwon, C. VanDam, S. E. Chiuve, H. W. Choi, P. Entler, P.-N. Tan, and J. Huh-Yoo, “Improving heart disease risk through quality-focused diet logging: Pre-post study of a diet quality tracking app,” JMIR Mhealth Uhealth, vol. 8, no. 12, p. e21733, Dec. 2020, doi: 10.2196/21733. "

- Choi, C. Chung, and H. Woo, “Diet-related mobile apps to promote healthy eating and proper nutrition: A content analysis and quality assessment,” Int. J. Environ. Res. Public Health, vol. 18, no. 7, p. 3496, Mar. 2021, doi: 10.3390/ijerph18073496.

- X.-T. Koo and K.-C. Khor, “Expense tracking with Tesseract optical character recognition v5: A mobile application development,” in Proc. 2023 IEEE Symp. Industrial Electronics & Applications (ISIEA), Kuala Lumpur, Malaysia, Jul. 2023, pp. 1–5, doi: 10.1109/ISIEA58478.2023.10212265.

THANK YOU